

Varunprasaath R. Selvaraju, Ph.D.

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PROFESSIONAL SUMMARY

PhD chemist/materials scientist with 6+ years in organic/materials synthesis, analytical characterization, and DOE-guided process development for data-rich molecular and polymer systems. Applies NMR, LC-MS/GC-MS, GPC/SEC, FT-IR, UV-Vis-NIR, TGA/DSC, and statistical process thinking to optimize compounds, formulations, and reproducible scale-up workflows.

CORE COMPETENCIES

Drug Delivery Chemistry

Organic Synthesis

High-Throughput Experimentation

Analytical Characterization

DOE/SPC

Process Scale-Up

Technical Documentation

WORK EXPERIENCE

University of Colorado Boulder

July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided chemical process development for bio-based polymer systems, defining critical process parameters, analytical readouts, and reproducible scale-up specifications.
- Translate material-quality and formulation/process data into concise technical reports, SOP-style documentation, and decision criteria for iterative experimental planning.
- Use spectroscopy, thermal analysis, and statistical design to connect processing conditions with material performance, stability, and manufacturability.

University of Colorado Boulder

July 2021 - July 2025

Graduate Researcher

- Designed, synthesized, purified, and characterized organic electrochromic molecules across 15+ variants, linking structure, purity, stability, and performance with NMR/GPC, MS, spectroscopy, thermal analysis, and electrochemistry.
- Built structure-property datasets to guide iterative molecule design, using analytical readouts to prioritize candidates and diagnose degradation or performance limitations.
- Produced peer-reviewed manuscripts, presentations, and technical summaries translating multi-technique characterization into actionable chemical-design recommendations.

Georgia Institute of Technology

Aug 2019 - July 2021

Graduate Researcher

- Drove process development for CVD-based molecular thin-film coatings, optimizing surface preparation and deposition conditions to improve reproducibility and optical performance.

ETH Zurich

May 2017 - July 2017

Research Assistant

- Optimized CVD polymer thin-film deposition under flow using DOE and in-situ QCM-D monitoring to quantify growth kinetics and process controls.

University of Colorado Boulder

2021 - 2025

Laboratory Operations Manager

- Managed characterization/process-development lab operations for a 15-member group; implemented calibration, SOP, and safety systems that improved instrument uptime by 30%.

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS**Bio-Based Polymer Scale-Up Process** Current

Built DOE-guided process windows, analytical readouts, and material-quality specifications for reproducible polymer/formulation scale-up.

DOE, SPC, process variables, analytical characterization, scale-up documentation

Electrochromic Molecular Design Chemistry of Materials 2025

Designed and evaluated molecular variants through synthesis, purification, analytical characterization, and property-driven iteration.

Organic synthesis, NMR, MS, FT-IR, UV-Vis-NIR, CV, structure-property analysis

AFOSR Defense Thin-Film Coatings DoD Funded

Converted early-stage coating concepts into controlled CVD process conditions and reproducible thin-film outputs.

CVD, QCM-D, surface prep, process optimization

EDUCATION**Ph.D. in Chemistry - University of Colorado Boulder**

2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials

Advisor: Prof. Seth Marder

(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc)

2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Synthesis & Drug-Delivery Chemistry: Organic synthesis, small-molecule/materials design, polymer/lipid-adjacent formulation concepts, purification, structure-property optimization

Analytical Characterization: NMR, LC-MS, GC-MS, GPC/SEC, FT-IR, UV-Vis-NIR, TGA, DSC, QCM-D, cyclic voltammetry

Process & Scale-Up: DOE, process windows, reproducibility studies, SOPs, scale-up specifications, CDMO-adjacent documentation

Data & Automation Readiness: SPC, JMP (basic), Python (basic), Excel, structured experimental records, technical reports

Materials Platforms: Bio-based polymers, electrochromic molecules, thin films, CVD coatings, surface functionalization

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." *Chemistry of Materials*, 2025, 37, 5558-5568.

2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." *RSC Advances*, 2018, 8, 20048-20055.

3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.

Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.

Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.

Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)